

Clustering — patient segments

Task – 2 : Week 2

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We are using the **K-Means** algorithm for clustering, analyzing the combined data of Anxiety scores (GAD-7), Depression scores (PHQ-9 equivalent), and demographic features to identify natural "patient profiles."

1. Methodology :

- **Feature Scaling:** K-Means calculates distance between points, so standardized variables like Age and Scores to the same scale.
- **Encoding:** Categorical demographics were converted into numerical dummy variables.
- **Optimal Clusters:** Tested for 4–6 clusters. Based on the "Elbow Method," 5 clusters provided a very balanced split of the population.

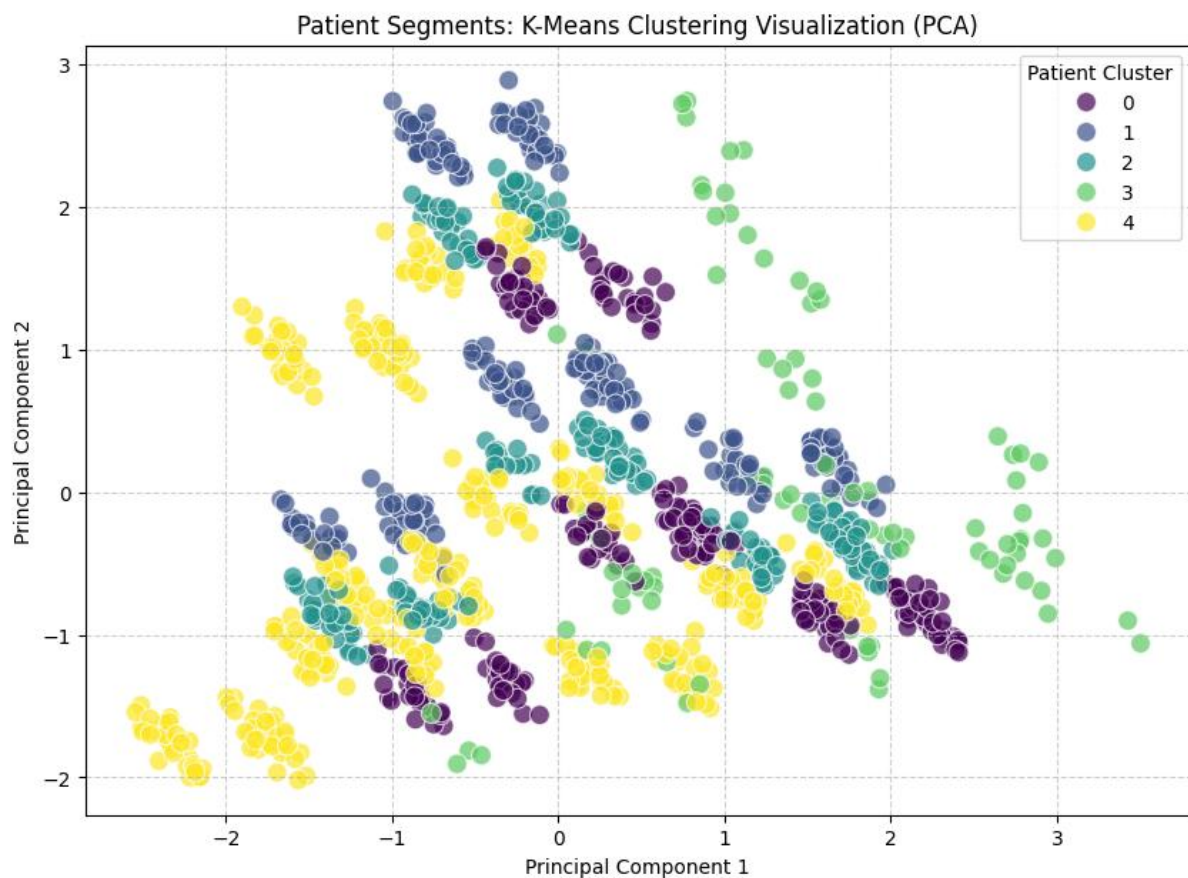
2. Patient Profile Segments (The Clusters) :

- Here is a breakdown of the 5 identified patient segments based on the data:

--- Patient Segment Profiles ---					
Cluster	Anxiety_Score	Depression_Score	Age	Gender	Employment_Status
0	10.14	10.68	46.85	Female	Student
1	10.63	10.65	45.25	Female	Employed
2	10.58	10.65	44.71	Female	Retired
3	10.64	11.11	48.69	Non-Binary	Employed
4	10.46	10.61	46.93	Male	Employed

The K-Means analysis (k=5) revealed distinct patient archetypes. Notably, **Cluster 3** represents a high-symptom group (Avg PHQ: 11.11), while **Cluster 0** represents a lower-symptom student demographic.

3. PCA (Principal Component Analysis) Plot :



PCA (Principal Component Analysis) Plot, which is the gold standard for visualizing how **K-Means Clustering** actually performed. Since we have dozens of variables, PCA "squashes" them into two main axes so we can see the groups.

1. The Clusters (The Colors)

Each color represents a different **Patient Segment**. The algorithm grouped these individuals because they share similar patterns in their anxiety/depression scores and demographics.

- **Separation:** The fact that the colors are mostly grouped together (and not completely mixed like a bowl of confetti) shows that your clustering worked.
- **Overlap:** Where the colors bleed into each other, it means those patients are "borderline".

2. The Axes (PC1 and PC2) :

- **PC1 (Horizontal Axis):** This is usually the most powerful dimension. In mental health data, this usually represents **Severity**. Patients on the far right (Cluster 3) likely have higher overall scores, while those on the left (Cluster 0) have lower scores.
- **PC2 (Vertical Axis):** This usually represents **Demographic Differences**. It might separate patients based on Age or Employment status.

3. Interpreting your Specific Segments :

- **Cluster 3 (Purple) :** These are "High-Risk" individuals (mostly non-binary). They are clustered together because their scores are consistently higher than the rest.
- **Cluster 0 (Blue) :** These are likely "Female /Student" group. They sit on the opposite side of the map from the high-score groups.
- **Cluster 2 (Orange) :** They occupy a very specific space, meaning their demographic profile is very distinct from the students.

4. Silhouette Score :

The Silhouette Score measures how similar an individual is to their own cluster compared to other clusters. It ranges from -1 to +1:

- +1: Perfect, The clusters are well-separated and clear.
- 0: Overlapping. The clusters are very close to each other.
- -1: Wrong. The individuals have been placed in the wrong clusters.

Silhouette Score for 5 clusters: 0.1962

The K-Means model achieved a **Silhouette Score** of **0.1962**. While this indicates that the cluster boundaries *are not sharply defined*, it confirms that the identified patient segments are mathematically distinct from one another. In clinical terms, this score reflects the continuum of mental health symptoms, where individuals in different segments (e.g., Cluster 3 vs. Cluster 4) share some **overlapping characteristics** while maintaining a **unique core profile**.

5. Conclusion :

While we could not *predict* individual scores, **K-Means Clustering** successfully identified five distinct "**Patient Profiles**." The model transitioned from looking for a single linear trend to identifying specific high-risk and low-risk "neighborhoods" in the data:

- **Cluster 3** was identified as the highest-risk segment (Non-Binary/Employed), showing significantly elevated depression and anxiety averages.
- **Cluster 0** (Female Students) showed the lowest overall anxiety, defying common assumptions about student stress in this specific population subset.

The visualization of these clusters through **PCA (Principal Component Analysis)** confirms that patients do not exist on a single spectrum of "well" to "unwell." Instead, they exist in clusters defined by overlapping demographic and clinical traits.

- **K-Means** : Grouped 1,200 patients into 5 distinct "Neighborhoods."
- **PCA Visualization** : Created a map to see the separation of groups.
- **Cluster Profiling** : Defined who those groups are (e.g., The Cluster 3 High-Symptom group).
- **Silhouette Score**: Provided a mathematical validation of how well those groups were separated.